

Device Access Control with AAA – Authentication, Authorization and Accounting



ICT2203 Networks Security

LAB 8 NOTES

2021-2022 Trimester 1

LEARNING OUTCOMES

In this lab you will learn ...

- **Authentication, Authorization and Accounting (AAA)**
 - a common security framework for mediating network and application access
- Management plane can be accessed via:
 - Console
 - Telnet
 - SSH
- Security of management plane is commonly achieved via Authentication, Authorization and Accounting (AAA).



Authentication - Who is allowed access?



Authorization - What are they allowed to do?



Accounting - What did they do?

In summary, here is a table comparing **TACACS+** vs **RADIUS** for implementing **AAA**.

	TACACS+	RADIUS
Functionality	Separates authentication, authorization and accounting, allowing modularity of the security server implementation	Combines authentication and authorization but separates accounting, allowing less flexibility in implementation than TACACS+.
Standard	Mostly Cisco supported	Open/RFC standard
Transport Protocol	TCP port 49	UDP port 1812 and 1813
Security	Entire packet encrypted	Password encrypted
Authorization	Provides authorization of commands on a per-user or per-group basis.	Has no option to authorize router commands on a per-user or per-group basis
Primary Use	Device administration	Network access

Lab Exercise 3

8.3

Accounting is to record what the user actually did.

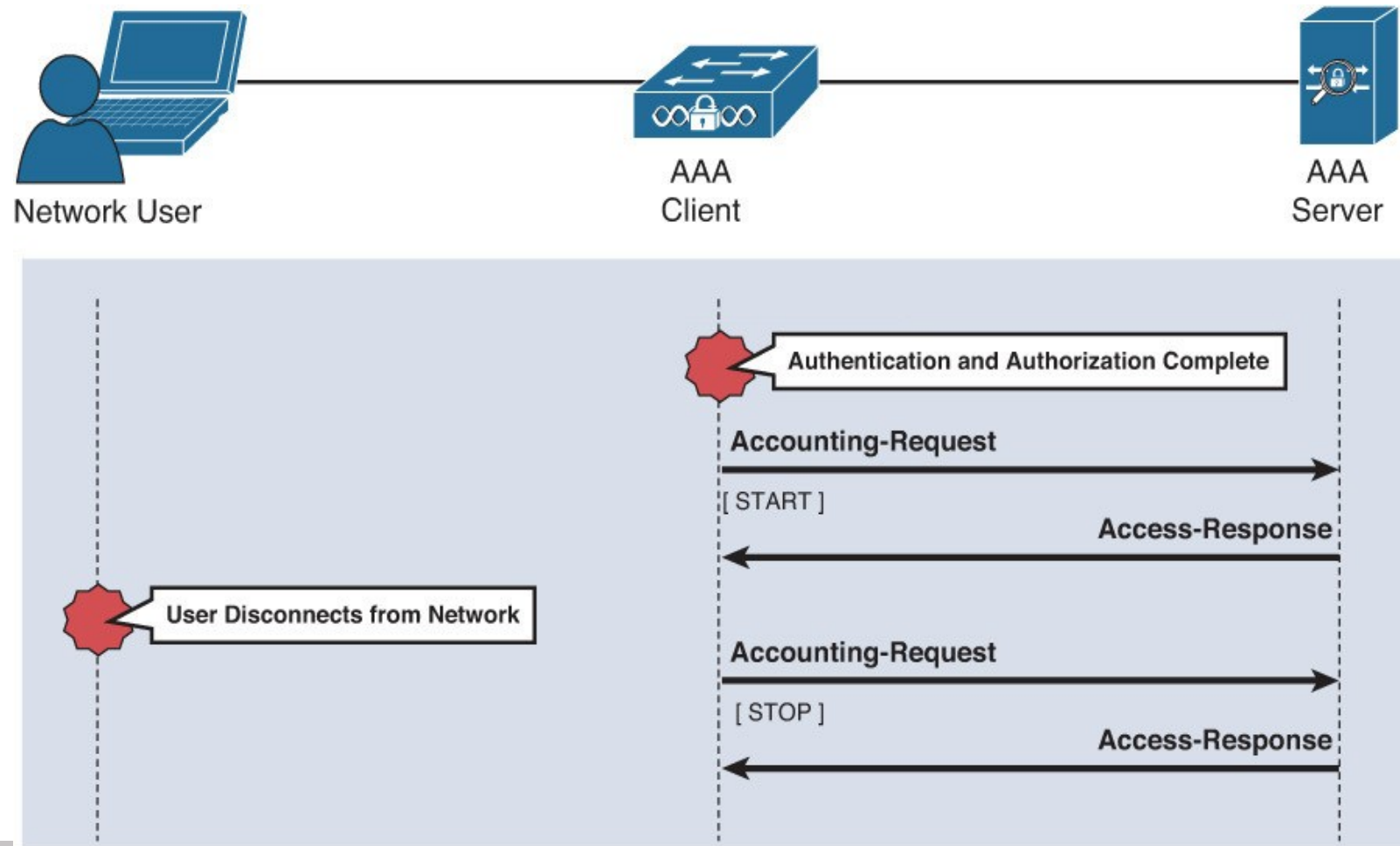
AAA accounting is to have a record of what a user has done which is also crucial to security.

One useful type of accounting is '**exec**' which provides information about user EXEC terminal session after login.

```
R1(config)# aaa new-model
R1(config)# ...
R1(config)# aaa authentication login default group tacacs+
group radiuslocal
R1(config)# aaa authorization exec default group tacacs+
group radiuslocal
R1(config)# aaa accounting exec default start-stop group
tacacs+ group radius
```

- **start-stop**: a **start** message is sent when session begins and a **stop** message is sent when session ends

For **RADIUS**, **accounting** is done separately after authentication and authorization are completed as shown:



Here's a sample of **exec log** in **freeRADIUS** server which can be found at `/var/log/freeradius/radacct/...`

```
Thu Nov  9 13:00:56 2017
```

```
Acct-Session-Id = "0000002F"
```

```
User-Name = "wendell"
```

```
Acct-Authentic = RADIUS
```

```
Acct-Status-Type = Start
```

```
NAS-Port = 1
```

```
NAS-Port-Id = "tty1"
```

```
NAS-Port-Type = Virtual
```

```
Service-Type = NAS-Prompt-User
```

```
NAS-IP-Address = 192.168.1.200
```

```
Acct-Delay-Time = 0
```

```
Event-Timestamp = "Nov  9 2017 13:00:56 UTC"
```

```
Tmp-String-9 = "ai:"
```

```
Acct-Unique-Session-Id = "f1e1339928a412881fcac9c442105e01"
```

```
Timestamp = 1510232456
```

```
Thu Nov  9 13:07:28 2017
```

```
Acct-Session-Id = "0000002F"
```

```
User-Name = "wendell"
```

```
Acct-Authentic = RADIUS
```

```
Acct-Terminate-Cause = User-Request
```

```
Acct-Session-Time = 392
```

```
Acct-Status-Type = Stop
```

```
NAS-Port = 1
```

```
NAS-Port-Id = "tty1"
```

```
NAS-Port-Type = Virtual
```

```
Service-Type = NAS-Prompt-User
```

```
NAS-IP-Address = 192.168.1.200
```

```
Acct-Delay-Time = 0
```

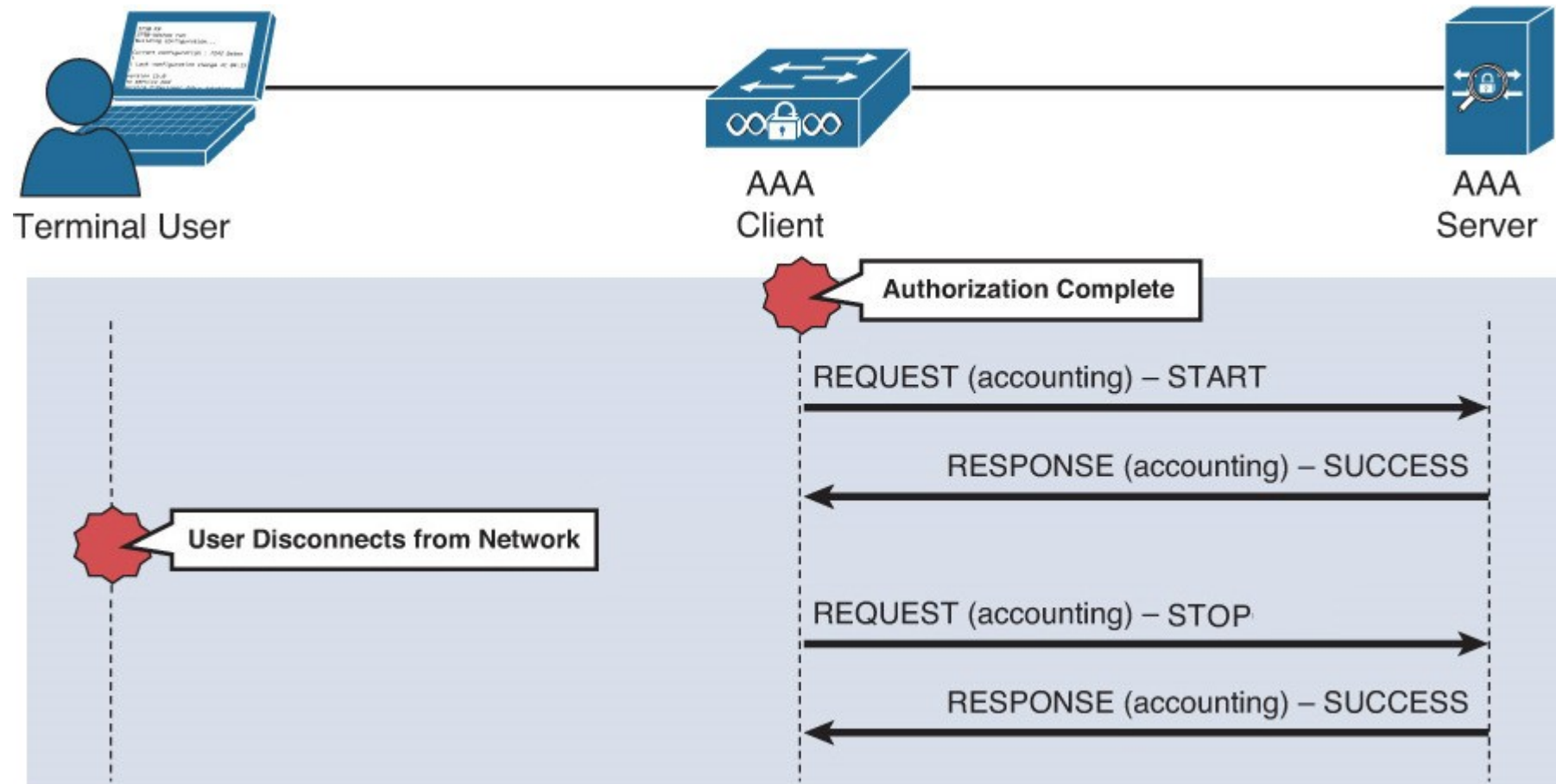
```
Event-Timestamp = "Nov  9 2017 13:07:28 UTC"
```

```
Tmp-String-9 = "ai:"
```

```
Acct-Unique-Session-Id = "f1e1339928a412881fcac9c442105e01"
```

```
Timestamp = 1510232848
```

Similarly for TACACS+, **accounting** is done separately after authorization as shown:



And here is a sample of **exec log** in **tacacs.net** server which can be found in

C:\Users\All Users\TACACS.net\Logs\Accounting_...

A '**start**' message is logged when the exec shell started:

```
<102> 2017-11-09 03:57:41 [192.168.1.200:43972]
11/09/2017 03:57:41 NAS_IP=192.168.1.200 Port=tty1
rem_addr=192.168.1.3 User=wendell Flags=Start task_id=25
timezone=UTC service=shell
```

A '**stop**' message is logged when the exec shell ended:

```
<102> 2017-11-09 03:58:04 [192.168.1.200:32751]
11/09/2017 03:58:04 NAS_IP=192.168.1.200 Port=tty1
rem_addr=192.168.1.3 User=wendell Flags=Stop task_id=25
timezone=UTC service=shell disc-cause=1 disc-cause-ext=9
pre-session-time=5 elapsed_time=22 stop_time=731022330
```

In addition, **TACACS+** (but not RADIUS) supports another useful type of accounting on **commands** which provide information about specific **privilege level commands** that a user issues.

```
R1(config)# aaa accounting exec default start-stop group  
tacacs+ group radius  
R1(config)# aaa accounting commands 15 default stop-only  
group tacacs+
```

- **stop-only**: a stop message is sent when command is executed

A sample of **commands log** in tacacs.net:

```
<102> 2017-11-09 03:57:47 [192.168.1.200:44251]  
11/09/2017 03:57:47 NAS_IP=192.168.1.200 Port=tty1  
rem_addr=192.168.1.3 User=wendell Flags=Stop task_id=25  
timezone=UTC service=shell priv-lvl=15 cmd=show  
running-config <cr>
```

```
R1(config)#enable secret level 15 ict2203
Router(config)#username admin privilege 15 secret letmein
Router(config)#aaa new-model
Router(config)#tacacs-server host 192.168.10.2 key tacacskey
Router(config)#radius-server host 192.168.10.10 key radiuskey
Router(config)#aaa authentication login default group tacacs+ enable
Router(config)#aaa authentication login mRADIUS group radius local enable
Router(config)#line vty 0 15
Router(config-line)#transport input ssh
Router(config-line)#login authentication mRADIUS
Router(config-line)#line con 0
Router(config-line)#login authentication default
Router(config-line)#end
```

```
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#test aaa group tacacs+ cisco cisco legacy

User successfully authenticated
```